AIPI 590 · EMERGING TRENDS IN EXPLAINABLE AI · PROJECT 1

When explanations disagree, which one do you trust?

A per-prediction **Explanation Trust Score** for post-hoc XAI — and a ground-truth benchmark testing whether disagreement actually signals that explanations are wrong.

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r=.61

Correlation between how much five explanation methods agree and how often they identify the **true** drivers — across 400 predictions.

1 The problem

Post-hoc explainers are how high-stakes model decisions get justified — a denied loan, a flagged tumour — and increasingly how organisations demonstrate compliance. But run several on the **same** prediction and they routinely disagree: different top features, different rankings, sometimes **opposite directions**. Krishna et al. (2022) named this the **disagreement problem** and found practitioners resolve it by intuition or by defaulting to a favourite tool.

Nobody is told **when** an explanation is unreliable. A single SHAP plot looks equally authoritative whether every method agrees with it or none do.

Research questions

- **Q1** Can disagreement be quantified into a usable per-prediction reliability signal?
- **Q2** When methods disagree, are explanations actually **wrong** more often?
- **Q3** What drives disagreement — the data, or the model?

The five methods audited

Shapley (sampling)	Game-theoretic credit over random feature orderings
LIME	Local weighted linear surrogate
Integrated Gradients	Path integral from an average input to this one
SmoothGrad × Input	Noise-averaged local gradient × input
Occlusion	Prediction drop when a feature is marginalised out

All five implemented from scratch in NumPy so they share one model, one background sample and one baseline — disagreement is methodological, not an artefact of library defaults. Verified exact on linear models; Shapley matches the reference `shap` library at r > 0.99.

2 One prediction, five stories

A single loan applicant, P(default) = 0.77, under a gradient-boosted model. Each column is a method; each cell is that method's **rank** for that feature; colour is the signed share of the explanation.

	Shapley	LIME	Integrated Gradients	SmoothGrad × Input	Occlusion
* debt_to_income	#1	#2	#1	#9	#1
* annual_income	#3	#1	#4	#1	#10
* credit_history_years	#8	#3	#2	#3	#6
* customer_id_hash	#2	#4	#6	#10	#2
* num_late_payments	#5	#7	#9	#7	#4
* monthly_spending	#6	#5	#3	#2	#5
* age	#9	#8	#5	#5	#7
years_at_address	#7	#9	#10	#8	#8
* loan_amount	#4	#6	#7	#4	#3

share of explanation (red → Default, blue → Repaid): 0.8, 0.6, 0.4, 0.2, 0.0, −0.2, −0.4, −0.6, −0.8

debt_to_income is ranked #1 by three methods and #9 by a fourth. **annual_income** is #1 for two and #10 for another. **customer_id_hash** — a meaningless identifier by construction — reaches the top 2 for two methods. Trust Score: **56 / 100 (Medium)**. * marks contested features: in some methods' top-5 but not others', or pushed in opposite directions.

3 What I built

1 Explain
5 methods · identical model, background and baseline

2 Compare
6 agreement metrics × 10 method pairs

3 Score
Trust Score 0–100 + contested features

4 Verify
Faithfulness tests + synthetic ground truth

$$\text{agree}(a,b) = \tfrac{1}{3} \cdot [\ \text{FeatureAgreement@}k + \text{SignAgreement@}k + (1 + \rho_{\text{Spearman}})/2\ ]$$

Trust = 100 × mean over 10 pairs

Which features matter, **which way** they push, and the **whole ranking**. Bands: High ≥ 70 · Medium · Low < 50. Metrics follow Krishna et al. (2022).

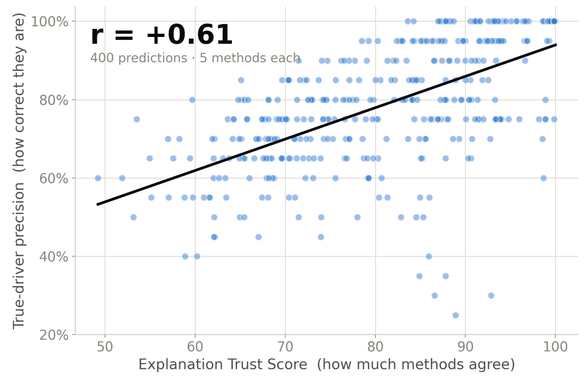
Verified two independent ways

- **Faithfulness** — PGI / PGU / deletion curves against a random-attribution control: does the explanation match the **model**?
- **Ground truth** — synthetic data whose true drivers are known by construction: does it match **reality**? 8 features, 4 informative, ρ swept 0 → 0.95, 4 model families, 20 predictions each.

Delivered artifact

6-page dashboard · REST API · CLI benchmark · CSV upload: audit your own model · 33 tests · Docker + CI · Methodology docs

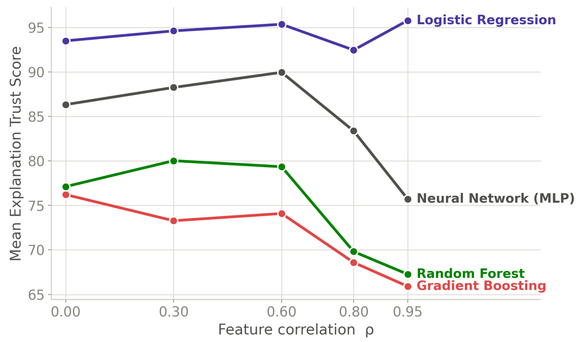
4 Finding 1 · Disagreement predicts error



Each point is one prediction: agreement among the five methods (x) versus the share of their top features that are genuinely informative (y).

The score earns its keep.
It is computed without any ground truth, yet tracks ground-truth correctness at **r = +0.61** — so an analyst can know when to distrust an explanation on real data, where truth is unavailable.

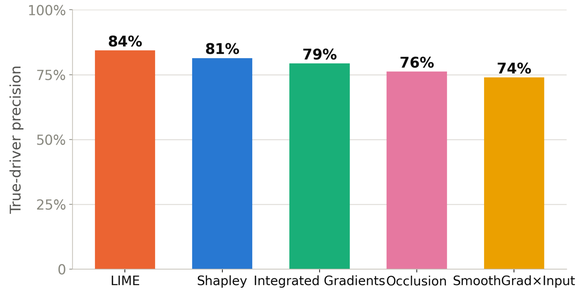
5 Finding 2 · Correlation erodes agreement



Mean Trust Score as pairwise feature correlation ρ rises from 0 to 0.95, by model family.

Interpretability is a property of the model, not only of the tool.
Tree ensembles and the neural net lose ≈ 10 points of agreement and ≈ 17 points of explanation accuracy. Logistic regression is unmoved (≈ 94) at every correlation level.

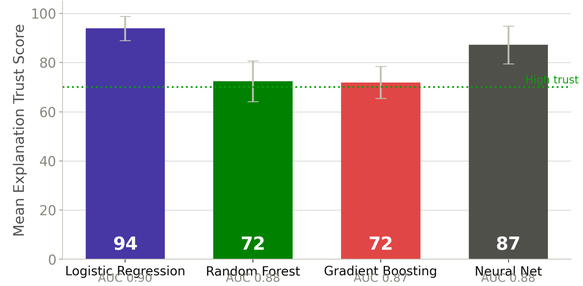
6 Finding 3 · The method matters



Share of each method's top-4 features that are genuine drivers, averaged over the benchmark.

Local gradients fail on trees; beware circular evaluation.
Tree ensembles are piecewise constant, so SmoothGrad's local gradient is near-zero and uninformative. Separately, Occlusion wins the removal-based faithfulness tests — but those tests **use Occlusion's own removal operation**. Against ground truth it is mid-table.

7 Same accuracy, different explainability



Credit-risk data, 30 predictions per model, ±1 SD. All four models sit within 0.03 AUC of each other, yet their explanations differ in reliability by more than 20 points.

A model-selection argument.
Accuracy cannot tell you how explainable a model will be. Where explanations carry legal or clinical weight, this makes the trade-off visible **before** deployment rather than after a contested decision.

2,000 explanations in the benchmark	**10** method pairs scored per prediction	**<0.5 s** all five methods, one prediction

8 Limitations & misuse risks

- **Agreement is not truth.** All five can agree and still be wrong — notably when the model genuinely relies on a proxy. The score is a warning light, not a certificate.
- **Circularity.** Removal-based faithfulness metrics favour removal-based explainers; reported openly inside the tool.
- **Off-manifold perturbation.** Under strong correlation every perturbation method queries the model on inputs that could not occur — part of **why** they diverge.
- **Explanation-washing.** The likeliest misuse is citing a high Trust Score as evidence that a model is fair or compliant. It certifies only that **methods concur**; it says nothing about accuracy, bias or appropriateness. It belongs **alongside** model validation, never instead of it.
- **Score-shopping.** k, the background sample or the method set could be tuned until the number flatters. Mitigations: fixed defaults, seeded runs, exported reports that record the configuration.
- **Scope.** Tabular binary classification; multi-class one-vs-rest. Bands (70 / 50) are reasoned, not yet calibrated.

9 Who it is for, and what is next

- **Data scientists** gain a reason to choose a method rather than a habit.
- **Auditors and compliance teams** gain a per-decision flag for "do not present a single explanation here".
- **Affected people** deserve a reason for a decision that does not depend on which library the analyst happened to import.

Future work

- **Calibrate the bands.** Fit Trust Score → probability-of-correctness by isotonic regression on the benchmark, turning a 0–100 heuristic into a stated error rate.
- **On-manifold explanations.** Conditional Shapley values, to test whether the correlation effect is an artefact of off-manifold sampling.
- **Stability as a sixth signal.** Re-run each method under different seeds to separate sampling noise from genuine methodological disagreement.
- **Human study.** Does showing a Trust Score change what practitioners do with an explanation? The problem was found through practitioner interviews; the remedy should be tested the same way.
- **Beyond tabular.** Text and vision models, where attribution disagreement is less studied and arguably more consequential.

REFERENCES

[1] Krishna, Han, Gu, Pombra, Jabbari, Wu & Lakkaraju (2022). *The Disagreement Problem in Explainable Machine Learning: A Practitioner's Perspective.* arXiv:2202.01602 · [2] Lundberg & Lee (2017). *A Unified Approach to Interpreting Model Predictions.* NeurIPS · [3] Štrumbelj & Kononenko (2014). *Explaining prediction models and individual predictions with feature contributions.* KAIS · [4] Ribeiro, Singh & Guestrin (2016). *"Why Should I Trust You?"* KDD · [5] Sundararajan, Taly & Yan (2017). *Axiomatic Attribution for Deep Networks.* ICML · [6] Smilkov et al. (2017). *SmoothGrad.* arXiv:1706.03825 · [7] Agarwal et al. (2022). *OpenXAI.* NeurIPS Datasets & Benchmarks · [8] Zeiler & Fergus (2014). *Visualizing and Understanding Convolutional Networks.* ECCV.

Data: seeded synthetic generators in-repo; UCI Breast Cancer Wisconsin and Efron et al. diabetes via scikit-learn. **AI use:** development assisted by Claude (Anthropic), per course policy.

RUN IT YOURSELF

`./run.sh` → dashboard
`./run.sh api` → REST API · `./run.sh test` → 33 tests
`./run.sh bench` → regenerate every number on this poster

Runs offline on a laptop. No GPU, no API keys.

Code, data and docs: `github.com/MAUZAMSHAFIBHAT/xai-disagreement-auditor`

Scan for code, docs and the full methodology